Year 13 Further Maths - Hyperbolic functions

Similar questions - Integration, inverse functions and volumes

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Show that $\operatorname{sech} x = \dfrac{2e^x}{e^{2x}+1}$. **[2]**

(b) Using the substitution $u = e^x$, find $\displaystyle\int \operatorname{sech} x \, dx$. **[4]**

(c) Hence find the exact value of $\displaystyle\int_0^{\ln\sqrt{3}} \operatorname{sech} x \, dx$. **[2]**

2

(a) Use the substitution $x = \sinh^2 u$ to find $\displaystyle\int \sqrt{\frac{x}{x+1}}\, dx$, giving your answer in the form $\sqrt{x(x+1)} - \sinh^{-1}(f(x)) + c$ where $f(x)$ is to be determined. **[6]**

(b) Hence find the exact area of the region between the curve $y = \sqrt{\dfrac{x}{x+1}}$, the x-axis and the lines $x = 1$ and $x = 2$. **[3]**

3

(a) Show that $\dfrac{d}{du}\left(\sinh^{-1} u\right) = \dfrac{1}{\sqrt{u^2+1}}$. **[2]**

(b) Find the equation of the normal to the curve $y = \sinh^{-1}(2x)$ at the point where $x = \sqrt{2}$, giving your answer in the form $y = mx + c$ with m and c in exact non-hyperbolic form. **[5]**

4

(a) Show that $\cosh(2\ln 2) = \dfrac{17}{8}$. **[2]**

(b) The region R is bounded by the curve $y = \sqrt{\sinh x}$, the x-axis and the line $x = 2\ln 2$. Find the exact volume of the solid formed when R is rotated through 2π about the x-axis. **[4]**

5 The curve C has equation $y = 16\cosh x - \sinh 2x$.

(a) Show that $\dfrac{d^2y}{dx^2} = 8\cosh x(2 - \sinh x)$. **[3]**

(b) Hence show that C has exactly one point of inflection and find its exact coordinates. **[4]**

6

(a) Express $17\cosh x - 15\sinh x$ in the form $e^{-x}\left(ae^{bx} + c\right)$ where a, b, c are integers. **[3]**

(b) The region bounded by $y = \dfrac{1}{\sqrt{17\cosh x - 15\sinh x}}$, the axes and the line $x = \ln 3$ is rotated through 2π about the x-axis. Use a suitable substitution and results from the formula book to show that the volume of the solid is $k\pi\tan^{-1} q$ where k and q are rational numbers to be determined. **[7]**

Question 1

(a) $\operatorname{sech} x = \dfrac{2}{e^x + e^{-x}} = \dfrac{2e^x}{e^{2x}+1}$

(b) $u = e^x$, $du = e^x dx$: $\displaystyle\int \frac{2}{u^2+1} du = 2\tan^{-1} u + c = 2\tan^{-1}(e^x) + c$

(c) $\left[2\tan^{-1} e^x\right]_0^{\ln\sqrt{3}} = 2\tan^{-1}\sqrt{3} - 2\tan^{-1} 1 = \dfrac{2\pi}{3} - \dfrac{\pi}{2} = \dfrac{\pi}{6}$

Question 2

(a) $dx = 2\sinh u \cosh u\, du$, $\sqrt{\dfrac{\sinh^2 u}{\sinh^2 u + 1}} = \tanh u$. Integral $= \int 2\sinh^2 u\, du = \int (\cosh 2u - 1)\, du = \frac{1}{2}\sinh 2u - u = \sinh u \cosh u - u$. With $\sinh u = \sqrt{x}$, $\cosh u = \sqrt{x+1}$: $\sqrt{x(x+1)} - \sinh^{-1}\sqrt{x} + c$, so $f(x) = \sqrt{x}$

(b) $\left[\sqrt{x(x+1)} - \sinh^{-1}\sqrt{x}\right]_1^2 = \left(\sqrt{6} - \sinh^{-1}\sqrt{2}\right) - \left(\sqrt{2} - \sinh^{-1} 1\right)$. $\sinh^{-1}\sqrt{2} = \ln(\sqrt{2}+\sqrt{3})$, $\sinh^{-1} 1 = \ln(1+\sqrt{2})$. Area $= \sqrt{6} - \sqrt{2} + \ln\dfrac{1+\sqrt{2}}{\sqrt{2}+\sqrt{3}}$

Question 3

(a) Let $y = \sinh^{-1} u$, $u = \sinh y$, $\dfrac{du}{dy} = \cosh y = \sqrt{1+\sinh^2 y} = \sqrt{1+u^2}$, so $\dfrac{dy}{du} = \dfrac{1}{\sqrt{u^2+1}}$

(b) $\dfrac{dy}{dx} = \dfrac{2}{\sqrt{4x^2+1}}$; at $x = \sqrt{2}$: $\dfrac{2}{3}$, so the normal gradient is $-\frac{3}{2}$. $y = \sinh^{-1}(2\sqrt{2}) = \ln(2\sqrt{2}+3) = 2\ln(1+\sqrt{2})$. Normal: $y - 2\ln(1+\sqrt{2}) = -\frac{3}{2}(x - \sqrt{2})$, i.e. $y = -\frac{3}{2}x + \frac{3\sqrt{2}}{2} + 2\ln(1+\sqrt{2})$

Question 4

(a) $\cosh(2\ln 2) = \frac{1}{2}(4 + \frac{1}{4}) = \frac{17}{8}$

(b) $V = \pi\int_0^{2\ln 2} \sinh x\, dx = \pi\left[\cosh x\right]_0^{2\ln 2} = \pi\left(\frac{17}{8} - 1\right) = \dfrac{9\pi}{8}$

Question 5

(a) $\dfrac{dy}{dx} = 16\sinh x - 2\cosh 2x$, $\dfrac{d^2y}{dx^2} = 16\cosh x - 4\sinh 2x = 16\cosh x - 8\sinh x\cosh x = 8\cosh x(2 - \sinh x)$

(b) $\cosh x > 0$, so $\dfrac{d^2y}{dx^2} = 0$ only when $\sinh x = 2$: $x = \sinh^{-1} 2 = \ln(2+\sqrt{5})$, a single solution, and $\dfrac{d^2y}{dx^2}$ changes sign there. $\cosh x = \sqrt{5}$, $\sinh 2x = 4\sqrt{5}$, so $y = 16\sqrt{5} - 4\sqrt{5} = 12\sqrt{5}$. Point: $\left(\ln(2+\sqrt{5}), 12\sqrt{5}\right)$

Question 6

(a) $17\cosh x - 15\sinh x = \frac{17-15}{2}e^x + \frac{17+15}{2}e^{-x} = e^x + 16e^{-x} = e^{-x}(e^{2x} + 16)$: $a = 1, b = 2, c = 16$

(b) $V = \pi\int_0^{\ln 3} \dfrac{e^x}{e^{2x}+16} dx$. Let $u = e^x$: $\pi\int_1^3 \dfrac{du}{u^2+16} = \pi\left[\frac{1}{4}\tan^{-1}\frac{u}{4}\right]_1^3 = \frac{\pi}{4}\left(\tan^{-1}\frac{3}{4} - \tan^{-1}\frac{1}{4}\right)$. Using $\tan^{-1} a - \tan^{-1} b = \tan^{-1}\dfrac{a-b}{1+ab}$: $\dfrac{3/4 - 1/4}{1 + 3/16} = \dfrac{8}{19}$. $V = \frac{1}{4}\pi\tan^{-1}\frac{8}{19}$, so $k = \frac{1}{4}$, $q = \frac{8}{19}$